

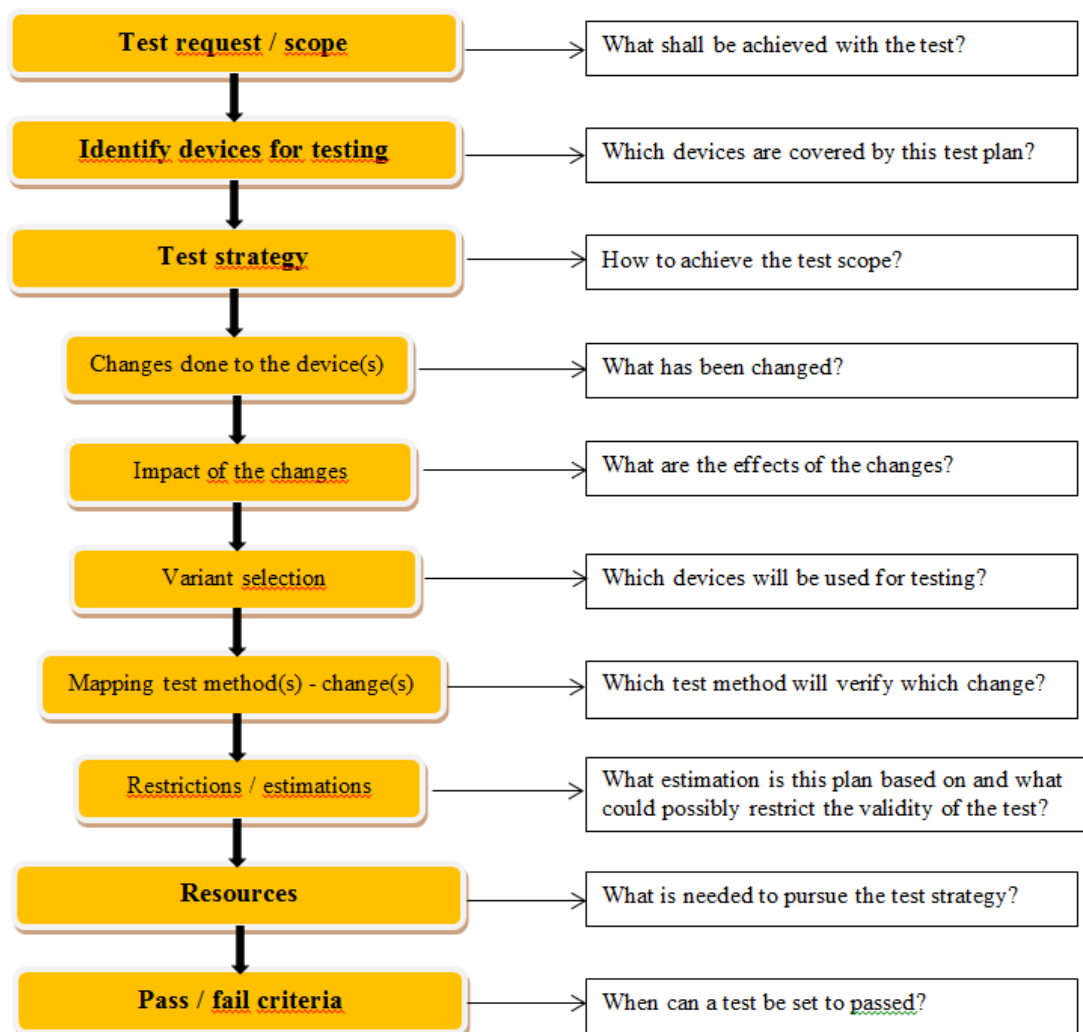
# Testplan

## 1 General explanation and tips regarding the test plan

See the following Wiki page for more information about test planning:

<https://wiki.ifm.com/pages/viewpage.action?pageId=9404793>

Short version:





## 1.1 Change history

(Type: e = created, m = modified, + = added, - = deleted)

| Version | Date       | Change by  | Type | Section | Description |
|---------|------------|------------|------|---------|-------------|
| V 0.1   | 11.04.2022 | M. Walther | e    |         |             |
|         |            |            |      |         |             |
|         |            |            |      |         |             |
|         |            |            |      |         |             |
|         |            |            |      |         |             |

Review minutes of the test plan: TBD



## 2 Scope

This document describes the type approval planning activities for project Cybersecurity.

It defines the objectives and scope of the verification activities.

The test plan is the plan for the procedure - it does not describe details about the test activities (which will be described in the test specification). This test plan does not contain any results, those will be documented in a test report.

### 2.1 Applicable documents

| R<br>ef | Document name | Date | Version |
|---------|---------------|------|---------|
| [1]     |               |      |         |
|         |               |      |         |
|         |               |      |         |
|         |               |      |         |

Table 2-1: Applicable documents

### 2.2 Abbreviations

| Abbreviation | Meaning                        |
|--------------|--------------------------------|
| BT           | BlueTooth                      |
| CCB          | Change Control Board           |
| DUT          | Device Under Test              |
| EMC          | Electro-Magnetic Compatibility |
| HW           | HardWare                       |
| IO           | Input/Output                   |
| IODD         | IO Device Description          |
| IOP          | InterOPerability               |
| LRD          | Line Recorder Device           |
| ME           | MEchanics                      |
| N/A          | Not Applicable                 |
| OS           | Operating System               |
| PLC          | Programmable Logic Controller  |
| Q2           | 2nd Quarter of the year        |
| QM           | Quality Management             |
| SVN          | SubVersioN                     |
| SW           | SoftWare                       |
| TA           | Type Approval                  |
| tbc          | To Be Confirmed                |
| tbd          | To Be Defined                  |
| TMG          | Technology Management Group    |



| Abbreviation | Meaning                   |
|--------------|---------------------------|
| UL           | Underwriters Laboratories |



### 3 Scope of the test activities

This document describes the type approval planning activities for project Cybersecurity. It defines the objectives and scope of the verification activities.

The test plan is the plan for the procedure - it does not describe details about the test activities (which will be described in the test specification). This test plan does not contain any results, those will be documented in a test report.

The plan specifies

- what must be tested
- why it must be tested (functional requirement, safety requirement, ...)
- how it will be tested (manually, automated, via review)
- who will test it (internally, externally)
- with what kind of measures it will be tested (interfaces, tools, ...)
- what resources are required (staff, material, tools)

The plan shall guide how to prove that the product fulfills the requirements.

#### 3.1 Test mission

☒ Verification if basic functionality of existing devices is not affected by attacks

☒ Preparation of use cases and Test Cases for new products

☒ Identification of countermeasure for new and existing devices

#### 3.2 Quality targets

When creating the test plan and the consecutive test specifications the following quality targets are to be taken into account:

☐ Low return rate

☐ Performance

☒ Robustness (operational)

☐ Precision

☐ Power

☒ Function

☐ Power consumption

☐ Compatibility

☐ Ergonomics

☒ Documentation

☐ IOP

☒ etc



### 3.3 Safety targets

Special focus in test planning shall be put on the verification of the following safety concepts and their implementation:

PLACEHOLDER currently no safety products affected

Note:

This is to be identified from the requirements for safety testing activities (see IEC 61508-2 / IEC 61508-3).

### 3.4 Development phases according V-model

| Test level verification                                            | Product | HW | SW | ME |
|--------------------------------------------------------------------|---------|----|----|----|
| <b>Type approval</b><br>(versus requirements)                      |         |    | X  |    |
| <b>System test / -integration</b><br>(versus system specification) |         |    | X  |    |
| <b>Module test</b><br>(versus design / architecture)               |         |    | X  |    |
| <b>Implementation / Development test</b>                           |         |    | X  |    |
| Please mark: <b>X</b> = Part of test plan                          |         |    |    |    |

Table 3-3: Mapping V-model to phase



## 4 Test object

This is the description of the product / component to be tested etc., as well as the assumptions and limitations that are relevant in terms of the characteristics, the content and scope of the test plan.

### 4.1 New devices related to this project

This test plan covers the products

| No | Component | HW | SW | Version | State |
|----|-----------|----|----|---------|-------|
| 1  |           |    |    |         |       |
| 2  |           |    |    |         |       |
| 3  |           |    |    |         |       |

Table 4-1: Identification of the test object



## 5 Test strategy

### 5.1 Test methods

Typical test methods are described in: [SEC-3 - Cybersecurity](#)

Typical exposed services or properties are described in: [SEC-105 - Attack Vector](#)

### 5.2 Change and impact analysis

#### 5.2.1 Changes to the DUT

There are no changes on the product.

#### 5.2.2 Impact of the changes

First overview / checklist for preselection (summary of the change and effect analysis):

| No. | Type of change | Physical change of behaviour | Possible impact | Type of test | Test-No.: |
|-----|----------------|------------------------------|-----------------|--------------|-----------|
| 1   | n/a            | n/a                          | n/a             | n/a          | n/a       |

**Table 5-1: Test effort**

### 5.3 Variant selection

PLACEHOLDER

### 5.4 Limitations and assumptions

The following limitations have been considered for the test strategy.

PLACEHOLDER

### 5.5 Measures to achieve the test targets

| No | Technology / Measure                         | Target to achieve                                                            | DUT's / variant |
|----|----------------------------------------------|------------------------------------------------------------------------------|-----------------|
| 1  | Functional Test Fieldbus Monitoring/Scanning | No influences in Fieldbus communication during and after test                |                 |
| 2  | Functional Test IoT Monitoring/Scanning      | Minor influence at IoT communication during test.<br>No influence after test |                 |
| 3  | Functional Test Fieldbus attacking           | No influences in Fieldbus communication during and after test                |                 |
| 4  | Functional Test IoT attacking                | Major influence at IoT communication during test.<br>No influence after test |                 |
| 5  |                                              |                                                                              |                 |

**Table 5-2: Test strategy**





## 6 Test documentation

### 6.1 V&V plan

The V&V plan is the upper level document for all V&V activities. It is the conjunction point for all test documentation. All lower level documents detail the information of the V&V plan.

See V&V plan: []

### 6.2 Test plan (this document)

The test plan describes the test activities for one development phase according to the V-model of ecomatic. The test plan is part of test documentation.

### 6.3 Test specifications

The test specification is the collection of all test cases / test case descriptions. There may be (e.g. subject-specific) different test specifications. The following are relevant for this test plan:

- **PLACEHOLDER**

#### 6.3.1 Test case description

Test cases have to be written in Polarion, which means that also the amount and selection of test cases is done in Polarion.

A typical test case description consists at least of the following:

- Start situation of test (including preparation)
- Test action
- Expected result

The test specification thus consists of a list (possibly hierarchically) of start situations, test actions and expected results as well as a statement, which of these actions should be executed to fulfill the testing strategy.

Before activation of a test case its testability must be shown. The test cases shall be linked to their appropriate requirements.

#### 6.3.2 Test protocol

If testing is planned based on a test specification, a Polarion test run with a unique name will be created in which the test results (and any observations etc.) will be documented.

The individual test cases can consist of 1 to n test steps. For each step the tester must give a verdict and for the whole test case a overall verdict. The following verdicts are possible:

##### **Pass:**

Test step or complete test case has the verdict "Pass". The results matches with the expectation.

##### **Fail:**

Test step or complete test case has the verdict "Fail". The result does not match with the expectation. If possible the defect ID shall be added to the comment field.

##### **Not planned:**

In test steps this verdict is used for steps that are not applicable. In such a case the test case shall be revised. Complete test cases will have this verdict if run in a wrong test run. In such cases comment shall be added to the comment field.

##### **Blocked**



Test step or complete test case could not be done (e.g. due to missing functionality). In such cases comment shall be added to the comment field.

## 6.4 Reports

### 6.4.1 Test reports

The results of the verification activities will be documented in a test report. The following test reports will be created and stored in the Polarion (70\_TypeTestReports) according QM-plan:

- **PLACEHOLDER**



## 7 Test resources

### 7.1 DUT type, amount and configurations

In order to carry out the type approval activities the following test equipment is required (see chapter "Test object").

| Article no. | Description | Version type | Amount |
|-------------|-------------|--------------|--------|
|             |             |              |        |
|             |             |              |        |
|             |             |              |        |

**Table 7-1: Test effort**

The detailed information about the used test equipment will be documented in the test run and/or the test report (serial number or similar identifier).

### 7.2 Effort

#### 7.2.1 Tasks to be done prior to testing

##### 7.2.1.1 Preparation of test set ups

PLACEHOLDER

Estimated effort: PLACEHOLDER

##### 7.2.1.2 Creation of test specifications

PLACEHOLDER

Estimated effort: PLACEHOLDER

##### 7.2.1.3 Further test planning, if needed

PLACEHOLDER

#### 7.2.2 Test execution type approval

| No | Technology / Measure | Expected number of TA test rounds | Effort with reporting (h) |
|----|----------------------|-----------------------------------|---------------------------|
| 1  |                      |                                   |                           |
| 2  |                      |                                   |                           |
| 3  |                      |                                   |                           |
| 4  |                      |                                   |                           |
| 5  |                      |                                   |                           |

**Table 7-2: Test effort**



### 7.3 Testers

Testing is planned to be done by the following persons:

| Role / Task | Assignees |
|-------------|-----------|
|             |           |
|             |           |
|             |           |
|             |           |
|             |           |

**Table 7-3: Resources**

### 7.4 Project schedule

The responsibility for creation of a master schedule for the activities derived from this test plan is held by the project management and is part of the project plan.

Type approval is expected to start **PLACEHOLDER**

### 7.5 Test environment

The test environment describes the facilities with the necessary equipment, tools, aids, etc. necessary to perform the tests in order to fulfill the tasks of the test plan and the related test specifications.

#### 7.5.1 Software / tools

For testing purposes the following software / tools are required:

| Designation | Version / Date |
|-------------|----------------|
|             |                |
|             |                |
|             |                |

**Table 7-4: Test effort**

#### 7.5.2 Hardware / test equipment

For testing purposes the following hardware / test equipment is required:

| Type | Designation | Serial number | calibrated |
|------|-------------|---------------|------------|
|      |             |               |            |
|      |             |               |            |
|      |             |               |            |

**Table 7-6: Test effort**



## 8 Assessment of test activities

### 8.1 Assessment of test activities

#### 8.1.1 Completeness

The evaluation of the testing activities includes an assessment of the completeness of the tests. Ideally, there shall be at least one test case per requirement.

#### 8.1.2 Condition of the test specification

The test specifications shall be written in Polarion. All test specifications must be reviewed and approved prior to start of testing. All testing shall be done according to the test specification. If the need for test case changes arises, their status shall be set back to draft when modified and a re-review shall take place (and the test specification shall also be approved again).

#### 8.1.3 Test environment

The following test environment shall be clearly defined in either the test specification (preferred) or the test run.

### 8.2 Assessment of the test results

#### 8.2.1 Pass/fail criteria

The "pass" verdict of the verification activities described in this test plan may be linked to the fulfillment of criteria that may be defined product and phase specific. If non-compliant the verification activities according to the test plan shall be considered as "fail" (exceptions: see below).

- **PLACEHOLDER**

*Remark: The "pass" and "fail" criteria are specified for the individual test case.*

### 8.3 Test planning risks

The following risks could affect or influence the test plan:

- **PLACEHOLDER**



## 9 Remarks

### 9.1 Traceability

Test cases versus requirements:

Traceability and the relationship of test cases to requirements is ensured by a reference to the unique ID.

- End of document -